

Stability analysis of a class of 2×2 triangular hyperbolic systems

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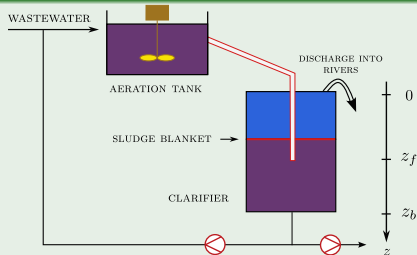
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LAGEP



Framework



1D-Dynamic Model of the Clarifier (Two-Phase flow)

State variables: ε_s (particles density), $f_s := \varepsilon_s v_s$ (momentum) with v_s the particles velocity

$$\begin{cases} \partial_t \varepsilon_s + \partial_z f_s = \frac{f_{1s}}{\rho_s} \delta_{z_f} \\ \partial_t f_s + \partial_z \left(\frac{f_s^2}{\varepsilon_s} + \frac{\sigma_e(\varepsilon_s)}{\rho_s} \right) = \underbrace{\varepsilon_s g \left(1 - \frac{\rho_l}{\rho_s} \right)}_{\text{Apparent weight}} + \underbrace{\frac{r(\varepsilon_s)(\varepsilon_s v_m - f_s)}{\rho_s \varepsilon_s (1 - \varepsilon_s)}}_{\text{Friction}} + \underbrace{\frac{f_{2s}}{\rho_s} \delta_{z_f}}_{\text{Feed}} \end{cases}$$

Two boundary conditions (**BCs**) at both ends of the spatial domain are needed to close the equations. These conditions are imposed by physical devices.

$$f_s(0, t) = U_0(t), \quad f_s(L, t) = U_1(t)$$

U_0, U_1 are *signal commands*.

—→ Coupling the equations with (**BCs**) corresponds to a *Boundary Control System*

Goal 1: Design a robust boundary controller that takes into account the Clarifier/Aeration Tank configuration (with disturbance inputs) in order to stabilize the sludge blanket depth

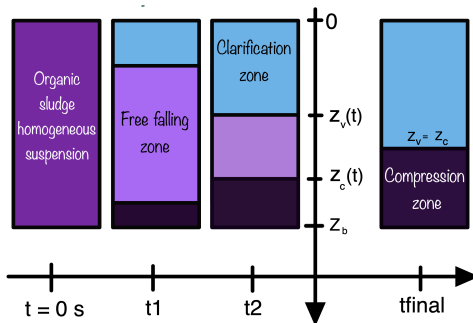
Goal 2: Prove the stability of the controlled system

OUTLINE OF TALK

- 1 BATCH COLUMN FRAMEWORK
- 2 THROUGH A SIMPLIFIED CLARIFIER
- 3 NUMERICAL RESULTS
- 4 CONCLUSIONS AND PERSPECTIVES

BATCH COLUMN FRAMEWORK

SETTLING IN A BATCH COLUMN



N. CHASSIN, C. VALENTIN, J.-M. CHOUBERT, F. COUENNE, C. JALLUT, 2022, *How to improve modeling of urban sludge settling in a batch column*, Note interne



C. VALENTIN, F. LAGOUTIÈRE, J.-M. CHOUBERT, F. COUENNE, C. JALLUT, 2023, *Knowledge-based model and simulations to support decision making in Wastewater Treatment*, Processes PROCEEDINGS 33rd European Symposium on Computer Aided Process Engineering (ESCAPE33), (6p), Athens. Greece

SYSTEM ANALYSIS

Governing equations

$D = [0, z_b] \times [0, T]$ / Find ε_s , f_s : $D \longrightarrow \mathbb{R}$ such that :

$$\begin{cases} \partial_t \varepsilon_s + \partial_z f_s = 0 \\ \partial_t f_s + \partial_z \left(\frac{f_s^2}{\varepsilon_s} + \frac{\sigma_e(\varepsilon_s)}{\rho_s} \right) = \varepsilon_s g \left(1 - \frac{\rho_l}{\rho_s} \right) - \tau \frac{\varepsilon_s^{\frac{n_r-1}{3-n_r}} f_s}{\rho_s (1 - \varepsilon_s)} \\ \varepsilon_s(z, 0) = \varepsilon_s^0, \quad f_s(z, 0) = f_s^0 \\ f_s(0, t) = 0, \quad f_s(z_b, t) = 0 \end{cases} \quad (1)$$

with: $\sigma_e(\varepsilon_s) = \begin{cases} 0, & \varepsilon_s \leq \varepsilon_c \\ \sigma_0 \frac{\varepsilon_s^{n_s} - \varepsilon_c^{n_s}}{\varepsilon_c^{n_s}}, & \varepsilon_s > \varepsilon_c \end{cases}$

ρ_l : water density ($:= 1000 \frac{\text{kg}}{\text{m}^3}$)

g : gravity acceleration constant ($:= 9.81 \frac{\text{m}}{\text{s}^2}$)

$\sigma_0, n_s, n_r, \varepsilon_c, \tau, \rho_s$ (parameters)

SYSTEM ANALYSIS

$$\underbrace{\partial_t \begin{pmatrix} \varepsilon_s \\ f_s \end{pmatrix}}_{\mathbf{Y}} + \underbrace{\partial_z \left(\frac{f_s^2}{\varepsilon_s} + \frac{\sigma_e(\varepsilon_s)}{\rho_s} \right)}_{\mathbf{F}(\mathbf{Y})} = \underbrace{\begin{pmatrix} 0 \\ \varepsilon_s g \left(1 - \frac{\rho_l}{\rho_s} \right) - \tau \frac{\varepsilon_s^{\frac{\eta_r-1}{3-\eta_r}} f_s}{\rho_s(1-\varepsilon_s)} \end{pmatrix}}_{\mathbf{S}(\mathbf{Y})}$$

Quasi-linear form :

$$\partial_t \mathbf{Y} + \mathbf{A}(\mathbf{Y}) \cdot \partial_z \mathbf{Y} = \mathbf{S}(\mathbf{Y})$$

where

$$\mathbf{A}(\mathbf{Y}) := \frac{\partial \mathbf{F}}{\partial \mathbf{Y}} \in \mathcal{M}_2(\mathbb{R})$$

Eigenvalues of $\mathbf{A}(\mathbf{Y})$?

$$\text{Sp}(\mathbf{A}) = \left\{ v_s - \sqrt{\frac{\sigma_0 n_s}{\rho_s \varepsilon_c} \varepsilon_s^{n_s-1}}, v_s + \sqrt{\frac{\sigma_0 n_s}{\rho_s \varepsilon_c} \varepsilon_s^{n_s-1}} \right\}, \text{ if } \varepsilon_s > \varepsilon_c$$

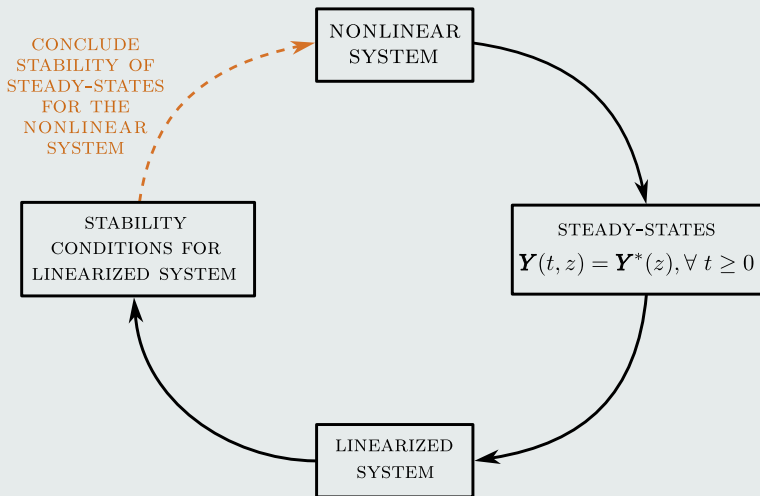
⇒ Strictly hyperbolic system

$$\text{Sp}(\mathbf{A}) = \{v_s, v_s\}, \text{ if } \varepsilon_s \leq \varepsilon_c$$

⇒ Weakly hyperbolic system

SYSTEM ANALYSIS

How to stabilize ?



SYSTEM ANALYSIS: STEADY-STATES

Solve the following system of ODEs:

$$\begin{cases} \frac{df_s^*}{dz} = 0, \\ \frac{d\left(\frac{f_s^{*2}}{\varepsilon_s^*} + \frac{\sigma_e(\varepsilon_s^*)}{\rho_s}\right)}{dz} = K\varepsilon_s^* - \gamma \frac{(\varepsilon_s^*)^\beta f_s^*}{1 - \varepsilon_s^*}, \\ f_s^*(0) = 0, \quad f_s^*(z_b) = 0 \end{cases}$$

RESULT

For any given initial condition $\mathbf{Y}^0$ and fixed set of parameters $(\sigma_0, n_s, n_r, \varepsilon_c, \tau, \rho_s)$, analytical expressions of system (1) steady-states are given by:

$$f_s^*(z) = 0, \quad \forall z \in [0, z_b] \quad (\text{I})$$

$$\varepsilon_s^*(z) = \begin{cases} 0, & 0 \leq z < z_c \\ (\varepsilon_c^{n_s-1} + (n_s - 1)\varepsilon_c^{n_s} g(\frac{\rho_s - \rho_l}{\sigma_0 n_s})(z - z_c))^{\frac{1}{n_s-1}}, & z_c \leq z \leq z_b \end{cases} \quad (\text{II})$$

NUMERICAL SIMULATION (GODUNOV-TYPE FINITE VOLUME SCHEME)

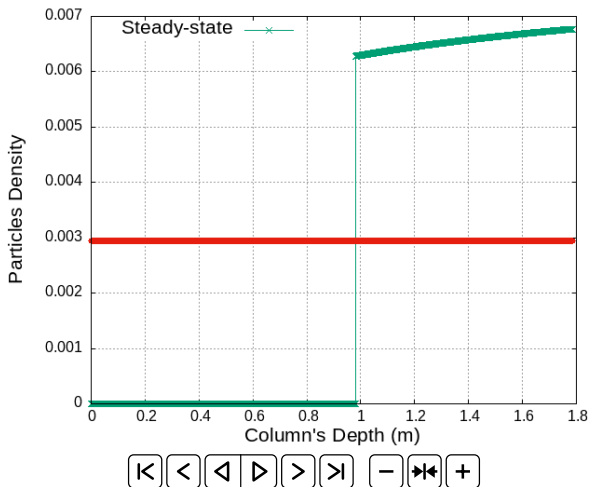


Fig. Simulation of the particles density on 6400 cells using Godunov scheme with an LLF Riemann Solver for flux computation: comparison with steady-state

SYSTEM ANALYSIS: LINEARIZATION

Consider deviations of the states with respect to steady-states:

$$E_s := \varepsilon_s(t, x) - \varepsilon_s^*(x), \quad V_s := v_s(t, x) - v_s^*(x)$$

Linearized system is written as:

$$\partial_t \tilde{\mathbf{Y}} + \mathbf{A}^*(z) \cdot \partial_z \tilde{\mathbf{Y}} + \mathbf{B}^*(z) \cdot \tilde{\mathbf{Y}} = 0 \quad (2)$$

$$\tilde{\mathbf{Y}} = \begin{pmatrix} E_s \\ V_s \end{pmatrix}, \quad \mathbf{A}^*(z) = \begin{pmatrix} v_s^* & \varepsilon_s^* \\ \frac{\sigma'_e(\varepsilon_s^*)}{\rho_s \varepsilon_s^*} & v_s^* \end{pmatrix}, \quad \mathbf{B}^*(z) = \begin{pmatrix} \text{Remaining} \\ \text{source terms} \end{pmatrix}$$

Completely decouple system (2) ?

Determine a transformation that diagonalizes the matrix $\mathbf{A}^*(z)$?

Not possible (due to weak-hyperbolicity)

Rather transform $\mathbf{A}^*(z)$ into a *Jordan* block:

$$\mathbf{A}^*(z) \longleftrightarrow \begin{pmatrix} c_1 & c_2 \\ 0 & c_3 \end{pmatrix}$$

THROUGH A SIMPLIFIED CLARIFIER

STABILITY ANALYSIS OF A SIMPLIFIED CLARIFIER: A TRIANGULAR HYPERBOLIC SYSTEM

Target System

$$\begin{cases} \partial_t \phi(t, x) = \Lambda(x) (A(x) \partial_x \phi(t, x) + B(x) \phi(t, x)), & x \in (0, 1), t \geq 0 \\ \phi(t, 1) = D \phi(t, 0), & \text{(BCs)} \\ \phi(0, x) = \phi^0(x), & \text{(ICs)} \end{cases} \quad (3)$$

$\phi(t, x) = (\phi_1(t, x), \phi_2(t, x))$ is the state,

$\Lambda(x) = \begin{bmatrix} \lambda_1(x) & 0 \\ 0 & \lambda_2(x) \end{bmatrix}$ satisfy $\lambda_{1,2}(x) > 0$ such that there exists some $x \in [0, 1]$

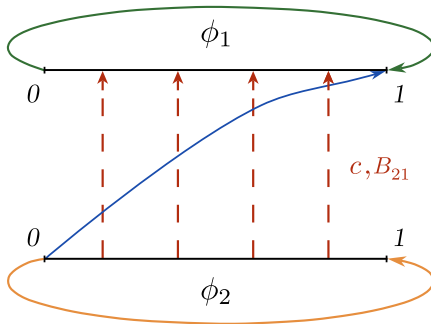
making $\lambda_1(x) = \lambda_2(x)$,

$A(x) = \begin{bmatrix} 1 & c(x) \\ 0 & 1 \end{bmatrix}$ where c is a continuous non-zero function,

$B(x) = \begin{bmatrix} B_{11}(x) & B_{12}(x) \\ 0 & B_{22}(x) \end{bmatrix}$ the source terms and $D = \begin{bmatrix} D_{11} & D_{12} \\ 0 & D_{22} \end{bmatrix}$ such that $|D_{11}|, |D_{22}| < 1$.

STABILITY ANALYSIS OF A SIMPLIFIED CLARIFIER: A TRIANGULAR HYPERBOLIC SYSTEM

$$\begin{cases} \partial_t \phi(t, x) = \Lambda(x) (A(x) \partial_x \phi(t, x) + B(x) \phi(t, x)), & x \in (0, 1), t \geq 0 \\ \phi(t, 1) = \begin{bmatrix} D_{11} & D_{12} \\ 0 & D_{22} \end{bmatrix} \phi(t, 0), \\ \phi(0, x) = \phi^0(x) \end{cases} \quad (4)$$



Schematic representation of the system (4)

STABILITY ANALYSIS OF A SIMPLIFIED CLARIFIER: A TRIANGULAR HYPERBOLIC SYSTEM

State space

$X = L^2((0, 1); \mathbb{R}) \times H^1((0, 1); \mathbb{R})$ equipped with the norm:

$$\|\cdot\|_X^2 : (f, g) \mapsto \|f\|_{L^2((0,1),\mathbb{R})}^2 + \|g\|_{L^2((0,1),\mathbb{R})}^2 + \|g'\|_{L^2((0,1),\mathbb{R})}^2$$

Exponential stability

The origin of system (2) is δ -exponentially stable if there exists an overshoot $\kappa \geq 1$ and a decay rate $\delta > 0$ such that for any initial condition $\phi^0 \in X$:

$$\|\phi(t; \phi^0)\|_X^2 \leq \kappa e^{-\delta t} \|\phi^0\|_X^2, \quad \forall t \geq 0 \quad (5)$$

Sufficient conditions that ensures exponential stability of this system ?

To answer, stability analysis is performed. The study will be divided into two cases: Without and with source term B

RESULTS: FIRST CASE $B = 0$

Recall:

$$D = \begin{bmatrix} D_{11} & D_{12} \\ 0 & D_{22} \end{bmatrix}, \quad \Lambda(x) = \begin{bmatrix} \lambda_1(x) & 0 \\ 0 & \lambda_2(x) \end{bmatrix}$$

Assumption 1

The function λ_2 and the matrix D satisfy the following inequality:

$$\max(D_{11}^2, D_{22}^2) < 1 - \frac{\overline{\lambda_2'}}{\underline{\lambda_2}} \quad (6)$$

Theorem 1

Let Assumption 1 holds and let $\nu := \frac{\overline{\lambda_2'}}{\underline{\lambda_2} - \lambda_2'}$, $\bar{\nu} := \frac{1 - D_{22}^2}{D_{22}^2}$.

Then, $\forall \delta < \min\left\{\frac{\bar{\nu}\lambda_1}{1+\bar{\nu}}, \frac{\bar{\nu}\lambda_2}{1+\bar{\nu}}, \frac{(\bar{\nu}-\nu)\lambda_2}{(1+\bar{\nu})(1+\nu)}\right\}$,

The origin of system (4) is δ -exponentially stable

RESULTS: SECOND CASE $B \neq 0$

Recall:

$$D = \begin{bmatrix} D_{11} & D_{12} \\ 0 & D_{22} \end{bmatrix}, \Lambda(x) = \begin{bmatrix} \lambda_1(x) & 0 \\ 0 & \lambda_2(x) \end{bmatrix}, B(x) = \begin{bmatrix} B_{11}(x) & B_{12}(x) \\ 0 & B_{22}(x) \end{bmatrix}$$

Assumption 2

The function λ_2 and the matrices B, D satisfy:

$$\max(D_{11}^2, D_{22}^2) < 1 - \max(2\overline{B_{11}}, \frac{\overline{\lambda_2'}}{\lambda_2} + 2\overline{B_{22}}) \quad (7)$$

Theorem 2

Let Assumption 2 holds and let

$$\nu_1 := 2\overline{B_{11}}(1 - 2\overline{B_{11}})^{-1}, \nu_2 := \frac{2\overline{\lambda_2'} + 2\lambda_2\overline{B_{22}}}{\lambda_2 - 2\overline{\lambda_2'} - 2\lambda_2\overline{B_{22}}}, \bar{\nu} := \frac{1 - D_{22}^2}{D_{22}^2}.$$

Then, $\forall \delta < \min\{\delta_1, \delta_2\}$ where $\delta_1 = \frac{\lambda_1(\bar{\nu} - \nu_1)}{(1 + \nu_1)(1 + \bar{\nu})}$ and $\delta_2 = \frac{\lambda_2(\bar{\nu} - \nu_2)}{(1 + \nu_2)(1 + \bar{\nu})}$, **the origin of system (4) is δ -exponentially stable.**

SKETCH OF THE PROOF:

The method used to prove the Theorems relies on the construction of a strict-Lyapunov functional (i.e. satisfying: $\dot{\mathcal{V}}(\phi) \leq -\delta \mathcal{V}(\phi)$).

Consider the following functional:

$$\mathcal{V}(\phi) = \mathcal{V}_1(\phi) + \alpha_{20} \mathcal{V}_{20}(\phi) + \alpha_{21} \mathcal{V}_{21}(\phi)$$

With

$$\mathcal{V}_1(\phi) = \int_0^1 \frac{1+\mu x}{\lambda_1(x)} \phi_1^2(x) dx,$$

$$\mathcal{V}_{20}(\phi) = \int_0^1 \frac{1+\mu x}{\lambda_2(x)} \phi_2^2(x) dx,$$

$$\mathcal{V}_{21}(\phi) = \int_0^1 \frac{1+\mu x}{\lambda_2(x)} (\partial_x \phi_2)^2(x) dx$$

and scalars $\mu, \alpha_{20}, \alpha_{21} > 0$.

Recall that $\mathcal{V}$ induces a norm on X that is equivalent to the state norm $\|\cdot\|_X$ ■

NUMERICAL RESULTS

EXAMPLE

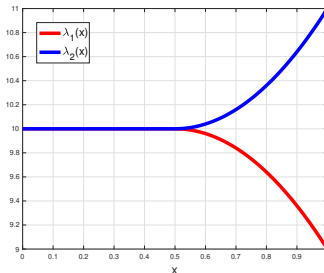
Consider system (4) with:

$$\lambda_1(x) = \begin{cases} 10, & x < \frac{1}{2} \\ -4(x - \frac{1}{2})^2 + 10 & \frac{1}{2} \leq x \leq 1 \end{cases}$$

$$\lambda_2(x) = \begin{cases} 10, & x < \frac{1}{2} \\ 4(x - \frac{1}{2})^2 + 10 & \frac{1}{2} \leq x \leq 1 \end{cases}$$

$$c(x) = x + 4, D = \begin{bmatrix} \frac{1}{2} & 1 \\ 0 & \frac{1}{2} \end{bmatrix}, B = \begin{bmatrix} \frac{x}{6} & 100x^3 \\ 0 & \frac{x}{6} \end{bmatrix}, \phi_1^0(x) = -\frac{2}{3},$$

$$\phi_2^0(x) = \frac{x^2}{2}(1 - \frac{x}{3}) - \frac{2}{3}.$$



ACADEMIC EXAMPLE

Regarding Theorem 1, the assumption $\max(D_{11}^2, D_{22}^2) = \frac{1}{4} < 1 - \frac{\lambda_2'}{\lambda_2} = \frac{3}{5}$ is satisfied.

Regarding Theorem 2, the assumption

$\max(D_{11}^2, D_{22}^2) = \frac{1}{4} < 1 - \max\left\{2\overline{B_{11}}, \frac{\overline{\lambda_2'}}{\lambda_2} + 2\overline{B_{22}}\right\} = \frac{4}{15}$ is satisfied.

The system is discretized using an *Upwind* type explicit finite difference scheme:

$$\begin{cases} \frac{\phi_{1,i}^{n+1} - \phi_{1,i}^n}{\Delta t} = \lambda_{1,i} \frac{\phi_{1,i+1}^n - \phi_{1,i}^n}{\Delta x} + c_i \lambda_{1,i} \frac{\phi_{2,i+1}^n - \phi_{2,i}^n}{\Delta x} \\ \frac{\phi_{2,i}^{n+1} - \phi_{2,i}^n}{\Delta t} = \lambda_{2,i} \frac{\phi_{2,i+1}^n - \phi_{2,i}^n}{\Delta x} \end{cases}$$

Characteristic lines
through (x,t) plane

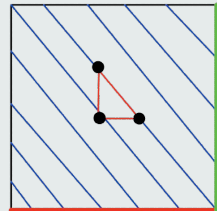
Setting:

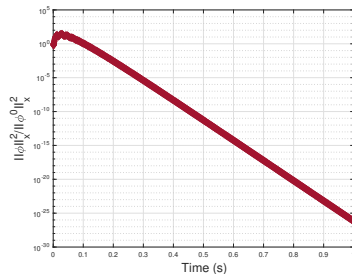
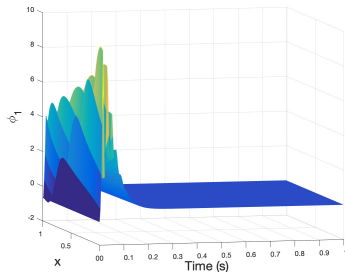
$$\gamma_{1,i} = \frac{\lambda_{1,i} \Delta t}{\Delta x}, \quad \gamma_{2,i} = \frac{\lambda_{2,i} \Delta t}{\Delta x}$$

finally we get the numerical scheme:

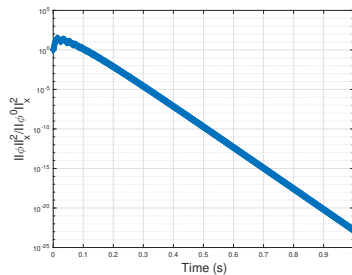
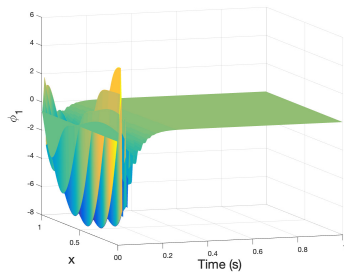
$$\phi_{1,i}^{n+1} = (1 - \gamma_{1,i}) \phi_{1,i}^n + \gamma_{1,i} \phi_{1,i+1}^n + c_i \gamma_{1,i} (\phi_{2,i+1}^n - \phi_{2,i}^n)$$

$$\phi_{2,i}^{n+1} = (1 - \gamma_{2,i}) \phi_{2,i}^n + \gamma_{2,i} \phi_{2,i+1}^n$$





System (2) without source term: (left) state ϕ_1 / (right) time evolution of $\frac{\|\phi\|_X^2}{\|\phi^0\|_X^2}$



System (2) with source term: (left) state ϕ_1 / (right) time evolution of $\frac{\|\phi\|_X^2}{\|\phi^0\|_X^2}$

EXAMPLE

Without source term, theoretically, decay rate is given by

$\min\left\{\frac{\bar{\nu}\lambda_1}{1+\bar{\nu}}, \frac{\bar{\nu}\lambda_2}{1+\bar{\nu}}, \frac{(\bar{\nu}-\nu)\lambda_2}{(1+\bar{\nu})(1+\nu)}\right\} = \frac{7}{2}$, whereas, numerically, decay rate is

$$\delta_{num} = 68.50 > \delta_{th}$$

With source term, decay rate is at least equal to $\delta_{th} = 0.16$, whereas, numerically, decay rate is:

$$\delta_{num} = 60.19 > \delta_{th}$$

The difference is due to the conservatism introduced by the proposed Lyapunov analysis

CONCLUSIONS AND PERSPECTIVES

CONCLUSIONS

Study of a class of Non-Strictly Hyperbolic problems:

- Tackling the *new* difficulties related to the non-diagonal structure
- Establishing sufficient stability conditions for this class of systems
- Lyapunov analysis with explicitly given overshoots and decay rates
- Numerical validation and comparison with theoretical results

Note that Non-Strictly Hyperbolic equations are of important interest and arise in many physical areas such as **Wastewater Treatment**, Elasticity theory, Advection of tracers in pipes etc.

PERSPECTIVES

Ongoing work

$$1) \quad \begin{cases} \partial_t \phi(t, x) = \Lambda(x) (A(x) \partial_x \phi(t, x) + B(x) \phi(t, x)), & x \in (0, 1), t \geq 0 \\ \phi(t, 1) = \begin{bmatrix} D_{11} & D_{12} \\ \mathbf{0} & D_{22} \end{bmatrix} \phi(t, 0), \\ \phi(0, x) = \phi^0(x) \end{cases}$$

Tackle the case where the system **is not in cascade** form (relax coefficient $\mathbf{D}_{21}$)

$\Rightarrow$ Design a wider class of controllers (of *Dynamical type* ?)

e.g.

$$\frac{d}{dt} \begin{bmatrix} \phi_1(t, 1) \\ \phi_2(t, 1) \end{bmatrix} = D \begin{bmatrix} \phi_1(t, 0) \\ \phi_2(t, 0) \end{bmatrix} + E \begin{bmatrix} \phi_1(t, 1) \\ \phi_2(t, 1) \end{bmatrix}$$

Where D and E are **full** matrices

2) Conclude about the stability of the nonlinear system

3) Apply this theory to the Clarifier/Aeration tank configuration.

Thank you for your attention
Questions ?